

Jamie Seoh

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Sep 2025 – Present

3.95/4.00 (91.2%) GPA

- Top 7.6% in Euclid Math Contest, 2 x Hackathon Winner, Software Engineering Entrance Scholarship

TECHNICAL SKILLS

Languages: C/C++, SystemVerilog, Python, Bash, RISC-V, MIPS, FPGA, Arduino, Teensy

Tools: Linux/Unix, Git/GitHub, Docker, Altera Quartus Prime, Fusion 360, Jira/Confluence, Figma

EXPERIENCE

AI Software Engineer Intern

Cerebras Systems

May 2026 – Aug 2026

Toronto, Ontario

- Engineered an end-to-end **hardware fault-injection tool** with C++ and ASM that simulates crashes, stalls, and slowdowns across the inference pipeline, helping engineers diagnose difficult-to-isolate customer failures.
- Reduced false-positive on-call alerts by **84%** through implementing a two-tier hardware-fault detection system.
- Built a **KV-cache breakpoint mechanism** that pauses faulted workloads and captures diagnostics for analysis.
- Authored deterministic **GTest** and **Pytest** tests to CI to catch regressions in new inference features and models.
- Sped up developer productivity with a static linter for Cerebras assembly language and an interactive dashboard.

Undergraduate Research Assistant

University of Waterloo

Sep 2026 – Present

Waterloo, ON

- Developing **SeaHorn**, an LLVM-based automated program verification under Professor Arie Gurfinkel.
- Investigating **Rust-to-LLVM** IR translation and verification semantics to extend tool support for Rust.

PROJECTS

Self-Playing MIDI Keyboard | C++, Arduino, Teensy 4.0

- Designed a 25-key MIDI keyboard with 3D-printed keys, velocity sensing, and USB-MIDI/synthesizer modes.
- Engineered a **Hall-effect sensing system** using multiplexed analog inputs, reducing pin usage from 25 to 6.
- Developed Teensy firmware for sensor processing, MIDI handling, velocity detection, and real-time key control.
- Built a **self-playing system** using solenoids, MOSFET switching, and PWM for independent key actuation.

Quota — DeltaHacks12 First Place | TypeScript, Next.js, LangChain, MongoDB

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed **10,000+**-line codebases in under **3 seconds** using **AST**-based inefficiency detection.
- Cut developer API spend by up to **40%** by providing real-time API cost insights via regex and usage caching.
- Created a React Flow sandbox for budget-aware system planning with **RAG Gemini** chatbot.

Nebula AI | React Native, FastAPI, PostgreSQL, Docker, GCP

- Shipped a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling text extraction and AI-driven summaries.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

COMMUNITY

Hack the North

Logistics Organizer

Feb 2026 – Present

Waterloo, Ontario

- Led end-to-end operations for **300+ volunteers** supporting Hack the North, a **1,000+**-participant hackathon.
- Redesigned the volunteer shift-selection system to prevent assignment race conditions and streamline signup.
- Oversaw **300+** 3D-printed keychains from design to production given time and material constraints.